

### **Description of Additional Supplementary Files**

**Title:** Supplementary Video 1

**Description:** Gut stimulus evoked activity across the brain, recorded using light-sheet microscopy in a fish expressing a genetically-encoded calcium indicator in almost all neurons. Trial-average data for one example fish is shown as top, side, and front maximum-intensity projections. Area Postrema, Parabrachial Nucleus, and Nodose ganglion are shown as zoom-ins. Video is 15x sped up. UV stimulus occurs near the start, indicated in the top left.